



Computing Infrastructures
April 10, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

!!! SOLUTIONS !!!

Exam Duration: 1hour

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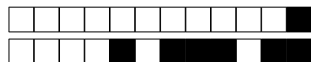
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The answers to the *Open Questions* should be written using ONLY the space available in the same page. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

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True false questions

Correct answer: +0.5, No answer: 0, Wrong Answer -0.25

Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 One advantage of building an in-house data center is that it grants an organization full autonomy over hardware choices and security policies. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 2 NVLink enables high-speed, direct communication among multiple GPUs, reducing reliance on CPU-based data transfer. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 3 Modern data centers often combine HDDs with SSDs to balance cost and performance. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 4 Warehouse-scale computers are typically composed of a large set of homogeneous servers. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 5 Edge computing nodes often cache and compute on locally produced data to reduce latency and congestion. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 6 Warehouse-scale computers are suitable only for limited workloads and cannot support large-scale services. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 7 SSDs have a limited number of read cycles, which can cause them to wear out over time. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 8 Tower servers are the dominant form factor for large data centers. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 9 The computing continuum spans from tiny IoT sensors all the way to large-scale data centers, enabling various levels of processing and storage. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 10 Building an in-house data center has no ongoing operational or maintenance costs once the facility is constructed. **SOLUTION: FALSE**

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. Write the final result on the last line of your answer (and its unit of measure if needed), preceded by the capitalized text "FINAL RESULT:". The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider an HDD with a data transfer time of 0.4ms for a 4 KB sector, a rotation speed of 12000 RPM, a mean seek time of 6 ms, and a negligible overhead of the controller. What is the average locality of the blocks if we observe an I/O service time of 40 ms to transfer a 120KB file?

!!! DRAFT OF THE SOLUTION !!!

12000RPM $\rightarrow$ Trot = 2.5ms

nBlocks = 120KB/4KB = 30

Tio = nBlocks*(1-L)*(Tseek+Trot) + nBlocks*Tt

40ms = 25*(1-L)*(6ms+3ms) + 25*0.4ms

L=89.02%



+1/4/57+

Question 12

Let us consider a set of requests in the disk queue referring to the following cylinders of the disk: 79, 14, 122, 65, 183, 37, 98. Consider the initial position of the disk head at cylinder 102, and it is moving from outside (higher cylinder number) to inside (lower cylinder number). If no further requests arrive, write the order of the served requests (from the first to the last) if the disk head scheduling algorithm adopted is SCAN. Use the cylinder number to refer to the request.

!!! DRAFT OF THE SOLUTION !!!

- 98 - 79 - 65 - 37 - 14 - 122 - 183 -



Question 13

Consider a system composed of three components: A, B, and C. The system fails whenever component A fails; otherwise, it remains operational as long as at least one of components B or C is functioning. The reliability of the system after five years is $R_{sys}(5y) = 0.68$, while the reliabilities of B and C after five years are $R_B(5y) = 0.6$ and $R_C(5y) = 0.5$, respectively. Failures are independent, and all components have exponentially distributed lifetimes. Compute the MTTF of component A.

!!! DRAFT OF THE SOLUTION !!!

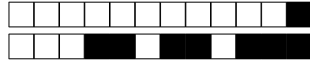
$$R_B C = 1 - (1 - R_B) * (1 - R_C)$$

$$R_A = R_{sys} / R_B C$$

$$R_A(5y) = e^{-t/MTTF_A}$$

$$MTTF_A = -t / \ln(R_A)$$

$$MTTF_A = 30.8y$$



Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in this page. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 14

⇒ Briefly describe the following infrastructure components used to organize multiple data centers located far apart, and explain the main differences among them: Geographic Areas, Computing Regions, and Availability Zones.

!!! EXAMPLE OF ANSWER !!!

A user request is fully processed within a single data center. However, multiple geographically distributed data centers are used to serve users across different parts of the world, to reduce latency, to increase throughput, and to improve availability. Geographic Areas, Computing Regions, and Availability Zones form a hierarchy used to organize distributed data center infrastructures.

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An Availability Zone (AZ) is the finest-grain location and exists inside a single computing region. AZs are fault-isolated locations with independent power, cooling, and networking. Their purpose is to support mission-critical applications with high availability and tolerance to data center failures. Unlike regions, AZs are close enough to support application-level synchronous replication, and typically at least three AZs are desirable to achieve quorum.

In summary, the main difference among them is at the level of the organization and resilience: Geographic Areas deal with geopolitical and data-residency boundaries, Computing Regions provide geographic separation and disaster recovery, and Availability Zones provide local fault isolation and high availability within a region.

Evaluation Grid

- 1 - Correct purpose of a distributed DC
- 3 - Definition of the hierarchy GA, CR, and AZ
- 1 - Clarity, wording/terminology, and organization of the answer



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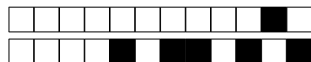
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True false questions

Correct answer: +0.5, No answer: 0, Wrong Answer -0.25

Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 SSDs have a limited number of read cycles, which can cause them to wear out over time. **SOLUTION: FALSE**

☐ A False

☐ B True

Question 2 Edge computing nodes often cache and compute on locally produced data to reduce latency and congestion. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 3 Modern data centers often combine HDDs with SSDs to balance cost and performance. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 4 NVLink enables high-speed, direct communication among multiple GPUs, reducing reliance on CPU-based data transfer. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 5 Building an in-house data center has no ongoing operational or maintenance costs once the facility is constructed. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 6 Warehouse-scale computers are typically composed of a large set of homogeneous servers. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 7 One advantage of building an in-house data center is that it grants an organization full autonomy over hardware choices and security policies. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 8 The computing continuum spans from tiny IoT sensors all the way to large-scale data centers, enabling various levels of processing and storage. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 9 Warehouse-scale computers are suitable only for limited workloads and cannot support large-scale services. **SOLUTION: FALSE**

☐ A False

☐ B True

Question 10 Tower servers are the dominant form factor for large data centers. **SOLUTION: FALSE**

☐ A True

☐ B False



+2/3/52+

Exercises

Correct answer: +2, No answer: 0.

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Consider an HDD with a data transfer time of 0.5ms for a 4 KB sector, a rotation speed of 12000 RPM, a mean seek time of 6 ms, and a negligible overhead of the controller. What is the average locality of the blocks if we observe an I/O service time of 30 ms to transfer a 100KB file?

!!! DRAFT OF THE SOLUTION !!!

12000RPM $\rightarrow$ Trot = 2.5ms

nBlocks = 100KB/4KB = 25

Tio = nBlocks*(1-L)*(Tseek+Trot) + nBlocks*Tt

30ms = 25*(1-L)*(6ms+3ms) + 25*0.5ms

L=91.76%



+2/4/51+

Question 12

Let us consider a set of requests in the disk queue referring to the following cylinders of the disk: 79, 14, 122, 65, 183, 37, 98. Consider the initial position of the disk head at cylinder 102, and it is moving from inside (lower cylinder number) to outside (higher cylinder number). If no further requests arrive, write the order of the served requests (from the first to the last) if the disk head scheduling algorithm adopted is SCAN. Use the cylinder number to refer to the request.

!!! DRAFT OF THE SOLUTION !!!

- 122 - 183 - 98 - 79 - 65 - 37 - 14 -



Question 13

Consider a system composed of three components: A, B, and C. The system fails whenever component A fails; otherwise, it remains operational as long as at least one of components B or C is functioning. The reliability of the system after five years is $R_{sys}(5y) = 0.72$, while the reliabilities of B and C after five years are $R_B(5y) = 0.6$ and $R_C(5y) = 0.5$, respectively. Failures are independent, and all components have exponentially distributed lifetimes. Compute the MTTF of component A.

!!! DRAFT OF THE SOLUTION !!!

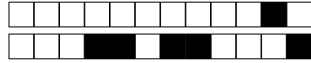
$$R_B C = 1 - (1 - R_B) * (1 - R_C)$$

$$R_A = R_{sys} / R_B C$$

$$R_A(5y) = e^{-t/MTTF_A}$$

$$MTTF_A = -t / \ln(R_A)$$

$$MTTF_A = 47.6y$$



Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in this page. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 14

⇒ Briefly describe the following infrastructure components used to organize multiple data centers located far apart, and explain the main differences among them: Geographic Areas, Computing Regions, and Availability Zones.

!!! EXAMPLE OF ANSWER !!!

A user request is fully processed within a single data center. However, multiple geographically distributed data centers are used to serve users across different parts of the world, to reduce latency, to increase throughput, and to improve availability. Geographic Areas, Computing Regions, and Availability Zones form a hierarchy used to organize distributed data center infrastructures.

A Geographic Area (GA) is the largest level. It is defined by geopolitical boundaries, such as country borders, and is mainly driven by data residency requirements and legal constraints on where data can be stored and processed. Each geographic area includes at least two computing regions.

A Computing Region (CR) is a smaller unit inside a geographic area and is the infrastructure granularity typically visible to customers. A CR may contain multiple data centers, but these are not exposed individually. Regions are separated by hundreds of miles and are placed in different risk areas, for example, different flood zones, so that large-scale disasters are less likely to affect them all at once. Because of this distance, synchronous replication across regions is generally not practical, while inter-region replication is suitable for disaster recovery.

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In summary, the main difference among them is at the level of the organization and resilience: Geographic Areas deal with geopolitical and data-residency boundaries, Computing Regions provide geographic separation and disaster recovery, and Availability Zones provide local fault isolation and high availability within a region.

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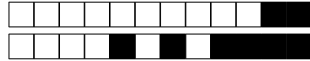
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True false questions

Correct answer: +0.5, No answer: 0, Wrong Answer -0.25

Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Tower servers are the dominant form factor for large data centers. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 2 Warehouse-scale computers are typically composed of a large set of homogeneous servers. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 3 NVLink enables high-speed, direct communication among multiple GPUs, reducing reliance on CPU-based data transfer. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 4 SSDs have a limited number of read cycles, which can cause them to wear out over time. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 5 Warehouse-scale computers are suitable only for limited workloads and cannot support large-scale services. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 6 Edge computing nodes often cache and compute on locally produced data to reduce latency and congestion. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 7 Modern data centers often combine HDDs with SSDs to balance cost and performance. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 8 Building an in-house data center has no ongoing operational or maintenance costs once the facility is constructed. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 9 One advantage of building an in-house data center is that it grants an organization full autonomy over hardware choices and security policies. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 10 The computing continuum spans from tiny IoT sensors all the way to large-scale data centers, enabling various levels of processing and storage. **SOLUTION: TRUE**

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

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!!! DRAFT OF THE SOLUTION !!!

10000RPM $\rightarrow$ Trot = 3ms

nBlocks = 100KB/4KB = 25

Tio = nBlocks*(1-L)*(Tseek+Trot) + nBlocks*Tt

30ms = 25*(1-L)*(6ms+3ms) + 25*0.5ms

L=92.22%



+3/4/45+

Question 12

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!!! DRAFT OF THE SOLUTION !!!

- 122 - 183 - 98 - 79 - 65 - 37 - 14 -



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Consider a system composed of three components: A, B, and C. The system fails whenever component A fails; otherwise, it remains operational as long as at least one of components B or C is functioning. The reliability of the system after five years is $R_{sys}(5y) = 0.68$, while the reliabilities of B and C after five years are $R_B(5y) = 0.6$ and $R_C(5y) = 0.5$, respectively. Failures are independent, and all components have exponentially distributed lifetimes. Compute the MTTF of component A.

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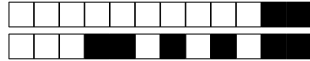
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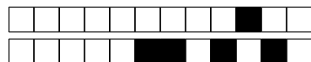
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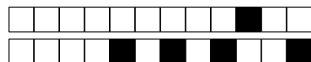
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True false questions

Correct answer: +0.5, No answer: 0, Wrong Answer -0.25

Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 One advantage of building an in-house data center is that it grants an organization full autonomy over hardware choices and security policies. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 2 Warehouse-scale computers are typically composed of a large set of homogeneous servers. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 3 Tower servers are the dominant form factor for large data centers. **SOLUTION: FALSE**

☐ A False

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Question 4 SSDs have a limited number of read cycles, which can cause them to wear out over time. **SOLUTION: FALSE**

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☐ A False

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Question 9 Edge computing nodes often cache and compute on locally produced data to reduce latency and congestion. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 10 Modern data centers often combine HDDs with SSDs to balance cost and performance. **SOLUTION: TRUE**

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

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!!! DRAFT OF THE SOLUTION !!!

10000RPM $\rightarrow$ Trot = 3ms

nBlocks = 100KB/4KB = 25

Tio = nBlocks*(1-L)*(Tseek+Trot) + nBlocks*Tt

30ms = 25*(1-L)*(6ms+3ms) + 25*0.5ms

L=92.22%



Question 12

Let us consider a set of requests in the disk queue referring to the following cylinders of the disk: 79, 14, 122, 65, 183, 37, 98. Consider the initial position of the disk head at cylinder 72, and it is moving from outside (higher cylinder number) to inside (lower cylinder number). If no further requests arrive, write the order of the served requests (from the first to the last) if the disk head scheduling algorithm adopted is SCAN. Use the cylinder number to refer to the request.

!!! DRAFT OF THE SOLUTION !!!

- 65 - 37 - 14 - 79 - 98 - 122 - 183 -



Question 13

Consider a system composed of three components: A, B, and C. The system fails whenever component A fails; otherwise, it remains operational as long as at least one of components B or C is functioning. The reliability of the system after five years is $R_{sys}(5y) = 0.68$, while the reliabilities of B and C after five years are $R_B(5y) = 0.6$ and $R_C(5y) = 0.5$, respectively. Failures are independent, and all components have exponentially distributed lifetimes. Compute the MTTF of component A.

!!! DRAFT OF THE SOLUTION !!!

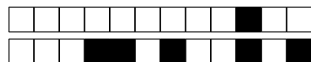
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$$MTTF_A = -t / \ln(R_A)$$

$$MTTF_A = 30.8y$$



Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

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Question 14

⇒ Briefly describe the following infrastructure components used to organize multiple data centers located far apart, and explain the main differences among them: Geographic Areas, Computing Regions, and Availability Zones.

!!! EXAMPLE OF ANSWER !!!

A user request is fully processed within a single data center. However, multiple geographically distributed data centers are used to serve users across different parts of the world, to reduce latency, to increase throughput, and to improve availability. Geographic Areas, Computing Regions, and Availability Zones form a hierarchy used to organize distributed data center infrastructures.

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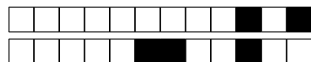
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In summary, the main difference among them is at the level of the organization and resilience: Geographic Areas deal with geopolitical and data-residency boundaries, Computing Regions provide geographic separation and disaster recovery, and Availability Zones provide local fault isolation and high availability within a region.

Evaluation Grid

- 1 - Correct purpose of a distributed DC
- 3 - Definition of the hierarchy GA, CR, and AZ
- 1 - Clarity, wording/terminology, and organization of the answer



Computing Infrastructures
April 10, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

!!! SOLUTIONS !!!

Exam Duration: 1hour

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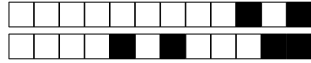
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Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. The numeric answer and its unit should be reported at the end of the exercise, preceded by the capitalized text "FINAL RESULT:".

The answers to the *Open Questions* should be written using ONLY the space available in the same page. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +0.5, No answer: 0, Wrong Answer -0.25

Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Warehouse-scale computers are typically composed of a large set of homogeneous servers. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 2 Tower servers are the dominant form factor for large data centers. **SOLUTION: FALSE**

☐ A False

☐ B True

Question 3 Modern data centers often combine HDDs with SSDs to balance cost and performance. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 4 Edge computing nodes often cache and compute on locally produced data to reduce latency and congestion. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 5 NVLink enables high-speed, direct communication among multiple GPUs, reducing reliance on CPU-based data transfer. **SOLUTION: TRUE**

☐ A True

☐ B False

Question 6 Building an in-house data center has no ongoing operational or maintenance costs once the facility is constructed. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 7 Warehouse-scale computers are suitable only for limited workloads and cannot support large-scale services. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 8 SSDs have a limited number of read cycles, which can cause them to wear out over time. **SOLUTION: FALSE**

☐ A True

☐ B False

Question 9 The computing continuum spans from tiny IoT sensors all the way to large-scale data centers, enabling various levels of processing and storage. **SOLUTION: TRUE**

☐ A False

☐ B True

Question 10 One advantage of building an in-house data center is that it grants an organization full autonomy over hardware choices and security policies. **SOLUTION: TRUE**

☐ A False

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Exercises

Correct answer: +2, No answer: 0.

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Question 11

Consider an HDD with a data transfer time of 0.4ms for a 4 KB sector, a rotation speed of 12000 RPM, a mean seek time of 6 ms, and a negligible overhead of the controller. What is the average locality of the blocks if we observe an I/O service time of 40 ms to transfer a 120KB file?

!!! DRAFT OF THE SOLUTION !!!

12000RPM $\rightarrow$ Trot = 2.5ms

nBlocks = 120KB/4KB = 30

Tio = nBlocks*(1-L)*(Tseek+Trot) + nBlocks*Tt

40ms = 25*(1-L)*(6ms+3ms) + 25*0.4ms

L=89.02%



+5/4/33+

Question 12

Let us consider a set of requests in the disk queue referring to the following cylinders of the disk: 79, 14, 122, 65, 183, 37, 98. Consider the initial position of the disk head at cylinder 72, and it is moving from outside (higher cylinder number) to inside (lower cylinder number). If no further requests arrive, write the order of the served requests (from the first to the last) if the disk head scheduling algorithm adopted is SCAN. Use the cylinder number to refer to the request.

!!! DRAFT OF THE SOLUTION !!!

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Consider a system composed of three components: A, B, and C. The system fails whenever component A fails; otherwise, it remains operational as long as at least one of components B or C is functioning. The reliability of the system after five years is $R_{sys}(5y) = 0.68$, while the reliabilities of B and C after five years are $R_B(5y) = 0.6$ and $R_C(5y) = 0.5$, respectively. Failures are independent, and all components have exponentially distributed lifetimes. Compute the MTTF of component A.

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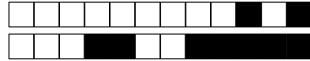
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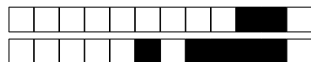
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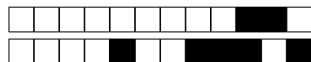
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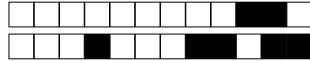
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+6/4/27+

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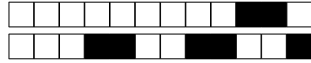
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